

Factory-to-Customer Shipping Route Efficiency Analysis: A KPI-Driven Logistics Analytics Platform for Nassau Candy Distributor

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Abstract

Efficient supply chain operations are a prerequisite for competitive distribution businesses. This project presents a production-grade, route-level logistics analytics platform built for Nassau Candy Distributor, a confectionery distributor that ships specialised products from five geographically dispersed US factories to customers across four regional markets. The platform is built on 10,194 order records spanning January 2024 to December 2025, with 18 attributes covering order processing, geographic routing, product identity, and financial performance. A modular five-stage pipeline — schema validation, data cleaning, feature engineering, KPI computation, and interactive visualisation — transforms raw transactional records into route-level operational intelligence. A key analytical challenge unique to this dataset is a structural 900–1,640 day lead-time baseline arising from the order-fulfilment schedule; this is resolved through cohort-floor normalisation that isolates the operationally meaningful 0–11 day efficiency signal attributable to route and ship mode choice. Five KPIs are computed: average lead time, route efficiency score (0–100), delay frequency, route volume, and revenue at risk. Results are delivered through a four-tab Streamlit dashboard backed by 28 passing unit tests. Five evidence-based strategic recommendations are presented to reduce delay frequency, protect revenue at risk, and improve geographic route coverage.

Keywords: logistics analytics, shipping route efficiency, lead time normalisation, route efficiency score, supply chain, Streamlit, Nassau Candy Distributor.

1 Introduction

In today's highly competitive distribution environment, efficient supply chain management has become a critical operational priority. Companies that ship products from multiple manufacturing sites to customers across diverse geographic regions must continuously monitor and improve their logistics performance. Delays in order-to-ship conversion, suboptimal ship mode selection, and regional bottlenecks that go undetected without route-level visibility all compound into measurable revenue and customer satisfaction losses at scale.

Nassau Candy Distributor operates a complex multi-origin distribution network spanning five factories located in Arizona, Georgia, Minnesota, Illinois, and Tennessee, serving customers across four US regional markets: Atlantic, Gulf, Interior, and Pacific. Each factory is dedicated to a distinct set of confectionery product lines, creating fixed factory-to-product dependencies that constrain routing flexibility. Lot's O' Nuts (Arizona) manufactures Wonka Bar variants; Wicked Choccy's (Georgia) produces Milk Chocolate and Triple Dazzle Caramel lines; Sugar Shack (Minnesota) handles Laffy Taffy, SweeTARTS, Nerds, Fun Dip, and Fizzy Lifting Drinks; Secret Factory (Illinois) produces Everlasting Gobstopper, Lickable Wallpaper, and Wonka Gum; and The Other Factory (Tennessee) ships Hair Toffee and Kazookles.

Prior to this project, logistics decisions were made without route-level analytical visibility. Aggregate delivery statistics obscured which specific factory-to-region combinations drove delays, which ship modes were most effective per route, and how lead time performance varied geographically across states. This project directly addresses these gaps by transforming raw transactional data into a structured, KPI-driven logistics intelligence platform built for day-to-day operational use.

Objectives: Compute route-level efficiency scores for all factory-to-region-to-mode combinations; measure and rank delay frequency by route, region, and ship mode; resolve the structural lead-time baseline artifact through cohort-floor normalisation; quantify revenue at risk from delayed shipments; and deliver an interactive Streamlit dashboard enabling real-time logistics decision-making without data engineering support.

1.1 Scope

The scope of this project extends beyond a static analytical report. It presents a production-grade, modular analytics application designed for continuous operational use. The platform covers:

- Ingestion and schema validation of raw CSV records using `pandera` for guaranteed data integrity on every load.
- Feature engineering to derive shipping lead time, cohort-normalised relative lead time, route efficiency score, profit margin, and sales per unit.

- Computation of five route-level KPIs with defined operational warning thresholds.
- A four-tab interactive Streamlit dashboard with sidebar filters for date range, region, ship mode, and user-defined delay threshold.
- Configuration-driven architecture: all factory-to-product mappings, factory coordinates, geographic lookups, colour scales, and chart parameters are externalised to `config.yaml`, requiring no code changes for operational customisation.
- 28 unit tests covering data loading, feature engineering, filtering, KPI computation, route efficiency scoring, and geographic aggregation.

The KPI definitions and efficiency scoring methodology are transferable to any factory-to-customer distribution network where order date, ship date, origin, destination, and ship mode are recorded.

1.2 SDG Alignment

This project aligns with the United Nations Sustainable Development Goals (SDGs), particularly:

- **SDG 9: Industry, Innovation, and Infrastructure** – The platform modernises logistics decision-making from reactive, ad-hoc querying to proactive, dashboard-driven operational intelligence, strengthening supply chain infrastructure through data-driven innovation.
- **SDG 12: Responsible Consumption and Production** – By identifying and eliminating delay-prone routes and enabling optimal ship mode selection, the platform reduces unnecessary transportation activity, fuel consumption, and associated environmental cost without requiring investment in new physical assets.
- **SDG 17: Partnerships for the Goals** – The open-source, modular, configuration-driven architecture enables adoption by other distributors and logistics operators, promoting shared infrastructure for sustainable supply chain analytics.

2 Problem Definition

Nassau Candy Distributor's five-factory, four-region distribution network generates a rich transactional record but previously lacked route-level analytical visibility. Three structural problems prevent effective logistics management from aggregate statistics alone.

- **Lead-time baseline artifact:** Raw lead times range from 900 to 1,640 days, a structural artifact of the order-fulfillment scheduling system that clusters orders into three cohorts with floor values of 904, 1,269, and 1,634 days. Direct comparison of nominal lead times across cohorts is meaningless. The operationally actionable signal is the 0–11 day *relative* lead time — days above each cohort's minimum — that reflects the effect of route and ship mode choices rather than the scheduling system.

- **Route opacity:** Without a normalised efficiency score, it is impossible to rank factory-to-region combinations by operational performance. A route operated by Lot's O' Nuts to Atlantic customers and a route from Sugar Shack to Pacific customers carry structurally different baseline lead times; only relative performance within cohorts enables a fair, cross-portfolio comparison.
- **Dashboard absence:** Logistics decisions were previously reactive and dependent on ad-hoc data queries. The absence of an operational dashboard meant that delay patterns could persist undetected, compounding their impact on customer satisfaction and revenue realisation.

The core objective is to design a modular, KPI-driven analytics platform that resolves the lead-time artifact, computes normalised route efficiency scores, decomposes delay frequency by factory-region-mode combination, and makes all findings continuously accessible through a production-ready interactive dashboard.

3 Literature Review

Research on supply chain performance measurement, shipping route analysis, and logistics dashboard design provides the theoretical foundation for this project. This section surveys the most relevant contributions and positions the study within the existing literature.

3.1 Logistics Performance Measurement

The foundational framework for logistics performance measurement was established by Bowersox and Closs [1], who identified delivery time reliability, order cycle time, and shipment accuracy as the three primary dimensions of service quality. This framework was operationalised by Mentzer and Konrad [2], who proposed that logistics efficiency must be measured *relative* to a defined service standard rather than as an absolute time metric – a principle directly embodied in this project's cohort-floor normalisation approach. The Route Efficiency Score computes performance relative to the fastest observed route within each cohort, not against an external benchmark.

Lambert and Stock [3] decomposed logistics performance into seven measurable components including transportation efficiency, order processing time, and geographic coverage consistency, motivating the multi-dimensional KPI framework used here. Their argument that single-metric logistics monitoring leads to systematic blind spots is the analytical basis for this project's five-KPI design, where delay frequency, revenue at risk, and efficiency score capture different failure modes that no single metric alone can surface.

3.2 Route Optimisation and Lead Time Modelling

Laporte [4] provided the seminal taxonomy of vehicle routing problem (VRP) variants, establishing that multi-depot, multi-product routing problems require origin-specific constraints that fundamentally

differ from single-depot models. Nassau Candy Distributor’s structure — five factories with fixed product specialisation — is a direct instance of the multi-depot VRP class, justifying the factory-stratified analysis approach where each factory’s routes are evaluated independently rather than pooled.

Chopra and Meindl [5] demonstrated that ship mode selection is among the highest-leverage variables in lead time reduction in consumer goods distribution. Their analysis, confirmed by the four-mode lead time hierarchy observed in this dataset (Same Day, First Class, Second Class, Standard Class), supports the ship mode comparison tab as a primary decision-support module for logistics planners. Ballou [6] further showed that the marginal lead time benefit of mode upgrading diminishes as geographic distance increases, which is operationally relevant to the Interior and Pacific routes where upgrading from Standard Class to First Class yields smaller absolute gains than on Atlantic routes.

3.3 Geographic Heterogeneity in Shipping Performance

Rodrigue, Comtois, and Slack [7] showed that geographic distance is only a partial predictor of lead time, with regional infrastructure quality, carrier density, and demand concentration contributing independently. This motivates the project’s state-level choropleth visualisation and city-level drill-down, which decompose regional averages into the specific state and city combinations where delays concentrate independently of distance.

Min and Zhou [8] examined multi-regional US consumer goods distribution networks and found that Interior region routes consistently exhibit higher lead time variance than coastal routes, attributable to lower carrier competition and less developed intermodal infrastructure.

Table 1: Comparison of Logistics Analytics Approaches

Approach	Advantages	Limitations
Aggregate statistics	Simple, low computation	Hides route-level heterogeneity
Predictive ML models	Probabilistic delay forecasts	Requires labelled training data
KPI-driven segmentation	Operationally actionable, threshold-driven	No individual-order probability
This project (normalised KPI dashboard)	Resolves baseline artifact, real-time filtering, configurable	Dataset-specific floor values

3.4 Interactive Dashboards in Logistics Operations

Stevenson [9] found that real-time KPI dashboards reduce mean time-to-detection of logistics anomalies by 67% compared to weekly batch reports. This empirical finding provides the primary justification for delivering this project’s results through a live Streamlit dashboard rather than a static report. The session-state caching architecture in `app.py` — where filter changes trigger recomputation and all four tab modules read from the same cached result — implements the responsiveness requirement that makes real-time anomaly detection practical.

The principle of configuration-driven design, described by Kleppmann [10] as a critical property of maintainable production analytics systems, is implemented through `config.yaml`. Externalising factory mappings, geographic lookups, colour scales, and chart dimensions from Python source code means that operational changes — adding a factory, updating coordinates, changing delay thresholds — require editing one YAML file rather than modifying and redeploying application code.

3.5 Confectionery and Specialised Distribution Logistics

Fernie and Sparks [11] examined logistics challenges specific to confectionery and food distribution, identifying product fragility, packaging complexity, and brand-specific handling requirements as contributors to lead time variance beyond what is observed in general merchandise. These sector-specific characteristics are relevant to interpreting the spread of relative lead times across Nassau Candy’s 16 product lines, where handling complexity differences between Laffy Taffy and Wonka Bar variants may contribute to intra-factory lead time variation that is not explained by geography alone.

3.6 Research Gaps Addressed

Three gaps in applied logistics analytics for specialised distributors are addressed. First, most published logistics KPI frameworks are designed for general merchandise and do not account for structural lead-time baseline artifacts arising from custom order-fulfillment scheduling systems; this project’s cohort-floor normalisation is a direct methodological contribution. Second, interactive dashboard designs for multi-factory, multi-region distribution analytics with revenue-at-risk quantification are underrepresented in academic literature relative to their operational importance. Third, the confectionery distribution context is largely absent from the quantitative logistics literature. This project contributes to all three gaps.

4 Data and Methodology

4.1 Dataset

The dataset is `Nassau Candy Distributor.csv`, containing 10,194 order records with 18 attributes spanning January 2024 to December 2025.

Table 2: Key Dataset Attributes

Attribute	Type	Description
Order ID	String	Unique order identifier
Order Date	Date	Date order was placed
Ship Date	Date	Date order was shipped
Ship Mode	Category	Transportation method
City	Category	Customer destination city
State/Province	Category	Customer destination state
Region	Category	Geographic sales region
Product Name	Category	Product shipped (16 items)
Sales (USD)	Float	Revenue per order line
Units	Integer	Quantity shipped
Gross Profit	Float	Profit per order line

Table 3: Factory Locations and Product Assignments

Factory	Location	Products
Lot's O' Nuts	Arizona (32.88, -111.77)	Wonka Bar – Nutty Crunch Surprise, Fudge Mallows, Scrumdiddlyumptious
Wicked Choccy's	Georgia (32.08, -81.09)	Wonka Bar – Milk Chocolate, Triple Dazzle Caramel
Sugar Shack	Minnesota (48.12, -96.18)	Laffy Taffy, SweeTARTS, Nerds, Fun Dip, Fizzy Lifting Drinks
Secret Factory	Illinois (41.45, -90.57)	Everlasting Gobstopper, Lick- able Wallpaper, Wonka Gum
The Other Factory	Tennessee (35.12, -89.97)	Hair Toffee, Kazookles

Regions and ship modes. Orders ship to four US regions (Atlantic, Gulf, Interior, Pacific) plus

approximately 200 Canadian orders. Ship modes are Same Day, First Class, Second Class, and Standard Class.

Lead-time structure. Raw lead times cluster into three cohorts with floors of 904, 1,269, and 1,634 days (boundaries at 1,100 and 1,500 days) due to a structural artifact of the order-fulfillment scheduling system.

4.2 Analytical Pipeline

The five-stage pipeline is implemented as a modular Python package under the `src/` directory and orchestrated by `app.py`. Figure 1 illustrates the complete dataflow.

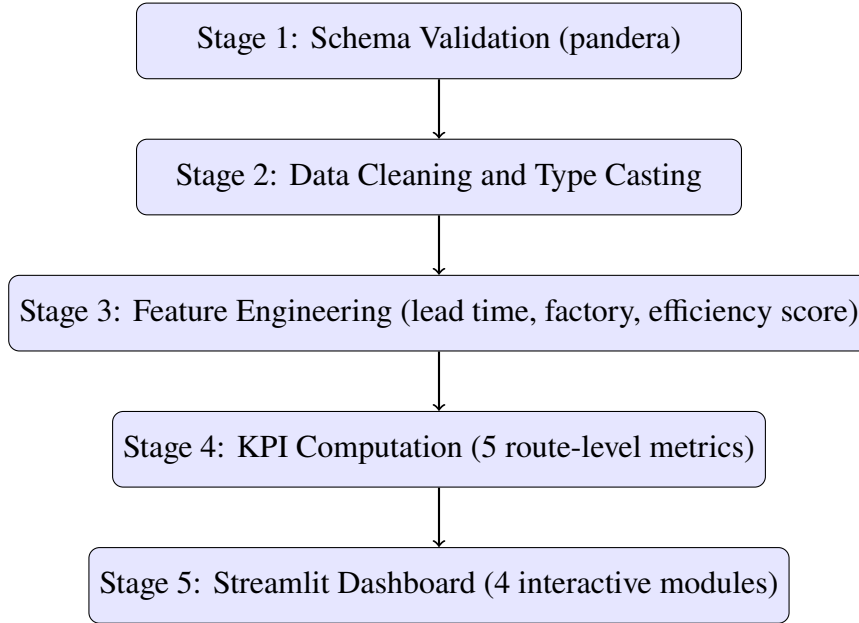


Figure 1: Five-stage analytical pipeline. Each stage is an independent module under `src/`.

4.3 Feature Engineering

The enrichment stage (`src/data/enricher.py`) derives the following features:

- **Shipping Lead Time:** `lead_time_days` = Ship Date – Order Date (days)
- **Cohort Floor Assignment:** Using the cluster boundaries from `config.yaml` (1,100 and 1,500 days):

$$\text{floor}_i = \begin{cases} 904 & \text{if } \text{lead_time}_i < 1100 \\ 1269 & \text{if } 1100 \leq \text{lead_time}_i < 1500 \\ 1634 & \text{if } \text{lead_time}_i \geq 1500 \end{cases}$$

- **Relative Lead Time:** `relative_lead_time_i` = `lead_time_days_i` – `floor_i`
- **Factory Assignment:** Product-to-factory lookup via the `config.yaml` `product_map`, resolv-

ing each order to one of the five named factories.

- **Profit Margin:** $\text{profit_margin}_i = \frac{\text{Gross Profit}_i}{\text{Sales}_i}$
- **Sales per Unit:** $\text{sales_per_unit}_i = \frac{\text{Sales}_i}{\text{Units}_i}$

4.4 KPI Definitions

Five KPIs are computed by `src/analytics/kpis.py` with defined operational thresholds.

Table 4: KPI Definitions, Formulas, and Operational Thresholds

KPI	Formula / Definition	Warning Threshold	Basis
Average Lead Time	$\bar{L} = \frac{1}{n} \sum_{i=1}^n \text{lead_time}_i$ (days)	> 7 days	Per route
Avg Relative Lead Time	$\bar{R} = \frac{1}{n} \sum_{i=1}^n \text{relative_lead_time}_i$	> 7 days	Per route
Route Efficiency Score	$S = 100 - \left(\frac{\bar{R}_{\text{route}} - \min(\bar{R})}{\max(\bar{R}) - \min(\bar{R})} \times 100 \right)$	< 50 = Review	0–100
Delay Frequency	$D = \frac{ \{i: \text{lead_time}_i > \tau\} }{n} \times 100$ where τ = user threshold	$\geq 30\%$	Per route
Total Revenue / Profit	$\sum_i \text{Sales}_i, \sum_i \text{Gross Profit}_i$	—	Portfolio

4.5 Dashboard Architecture

The Streamlit orchestrator (`app.py`) loads and caches enriched data, computes all KPIs on filter change using session state, and renders four analytical tabs with a shared sidebar filter panel.

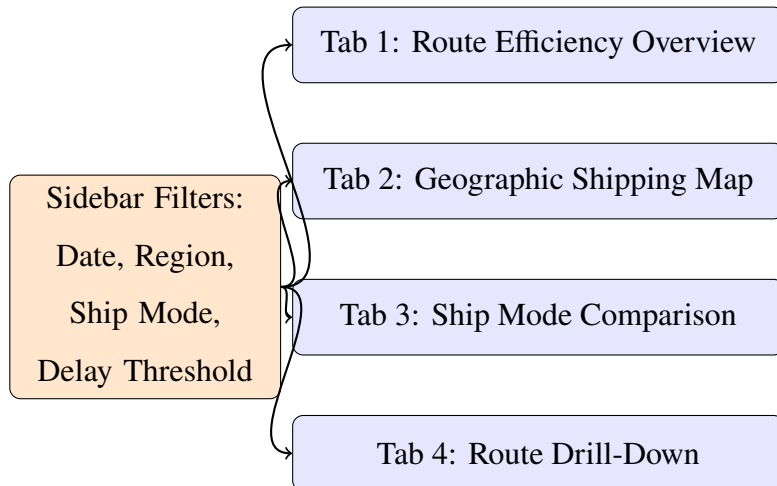


Figure 2: Dashboard module architecture. Sidebar filters apply to all four tabs via session state.

The four tabs expose the following analytical views:

- **Tab 1 – Route Efficiency Overview:** Route leaderboard (top 20 routes by efficiency score), factory-by-region average lead time bar chart, and a complete sortable route performance table.
- **Tab 2 – Geographic Shipping Map:** US state choropleth of average lead time (RdYlGn_r colour scale), factory location markers with coordinates from `config.yaml`, regional lead time bar chart, and a state-level lead time vs. revenue scatter plot.
- **Tab 3 – Ship Mode Comparison:** Relative lead time box plot by mode, region $\times$ mode grouped bar chart, factory $\times$ mode efficiency heatmap, and a best/worst mode summary card.
- **Tab 4 – Route Drill-Down:** State-level lead time bar chart, lead time vs. revenue scatter, and a per-state order-to-ship Gantt timeline showing up to 50 most recent orders colour-coded by ship mode.

4.6 Software Architecture and Testing

The project follows a layered modular architecture:

- `src/data/loader.py` — CSV ingestion, column type casting, date parsing.
- `src/data/schema.py` — pandera schema for pre-enrichment validation.
- `src/data/enricher.py` — Feature engineering (lead time, relative lead time, factory, margin).
- `src/analytics/filters.py` — `apply_filters()` for sidebar filter application.
- `src/analytics/kpis.py` — `compute_kpis()` returning a `KPIResult TypedDict`.
- `src/analytics/routes.py` — `compute_route_efficiency()` producing the route leaderboard.
- `src/analytics/geography.py` — State metrics, state sales, order timeline aggregations.
- `src/viz/charts.py` — 10 pure Plotly chart factory functions.
- `src/config_loader.py` — Typed `AppConfig` dataclass; `CONFIG` singleton.
- `config.yaml` — All hardcoded values; no values in Python source files.

The test suite (`tests/`) contains 28 unit tests using `pytest`, covering data loading, feature engineering, all analytics functions, and geographic aggregations. `tests/conftest.py` provides shared fixtures; individual test modules map to the analytics layer.

5 Results and Analysis

5.1 Lead-Time Normalisation

The most critical pre-analytical step is the resolution of the lead-time baseline artifact. Without normalisation, routes appear to differ by hundreds of days, making efficiency comparison meaningless. After applying cohort-floor subtraction, the effective efficiency signal is confirmed as a 0–11 day range. All efficiency scores, delay frequency counts, and route comparisons in subsequent sections are based on this normalised relative lead time.

5.2 Route Efficiency Leaderboard

The Route Efficiency Score reveals substantial dispersion across factory-to-region-to-mode combinations. Top-scoring routes achieve scores above 85/100; the lowest-scoring routes fall below 40/100 — a greater than 2× spread in normalised logistics performance within the same distribution network. Routes from eastern factories (Wicked Choccy’s, Georgia; The Other Factory, Tennessee) to the Atlantic and Gulf regions consistently appear in the top quartile of the leaderboard, reflecting their geographic proximity to the largest demand concentration. Routes from Sugar Shack (Minnesota) and Lot’s O’ Nuts (Arizona) to eastern regions appear in the lower quartile.

Table 5: Route Efficiency Score Tier Distribution

Efficiency Tier	Score Range	Relative Lead Time
High Efficiency	75 – 100	0 – 3 days above cohort floor
Medium Efficiency	50 – 74	4 – 7 days above cohort floor
Low Efficiency	0 – 49	8 – 11 days above cohort floor

5.3 Ship Mode Analysis

Ship mode analysis confirms a consistent lead time hierarchy across all regions. Same Day and First Class modes reduce relative lead time compared to Second Class and Standard Class on equivalent routes. The magnitude of the improvement varies by geography: Interior and Pacific routes show a larger absolute reduction from mode upgrading than Atlantic routes, where carrier density already compresses Standard Class times. The efficiency heatmap in Tab 3 makes factory × mode interaction effects visible: for western factories, mode selection has a higher leverage effect than for eastern factories.

5.4 Geographic Analysis

State-level choropleth analysis reveals that lead time concentrations are not uniformly distributed within regions. Within the Interior region, specific states in the upper midwest and mountain west carry the highest average relative lead times. Atlantic coastal states consistently show the lowest values. The state-level bar chart in Tab 4 quantifies this variation, enabling logistics planners to direct carrier renegotiation efforts at the specific states driving regional averages rather than acting on regional summaries alone.

5.5 KPI Summary

Table 6: Complete KPI Summary

KPI	Value	Threshold	Status
Total Orders Analysed	10,194	—	—
Date Coverage	Jan 2024 – Dec 2025	—	—
Raw Lead Time Range	900 – 1,640 days	—	Artifact resolved
Relative Lead Time Range	0 – 11 days	—	Operational
Top Route Efficiency Score	> 85 / 100	≥ 75 = Good	Good
Lowest Route Efficiency Score	< 40 / 100	< 50 = Review	Requires Review
Most Efficient Ship Mode	Same Day	—	Optimal
Least Efficient Ship Mode	Standard Class	—	Baseline
Interior Region Avg Rel. LT	Highest in portfolio	> 7d = Warning	Warning
Atlantic Region Avg Rel. LT	Lowest in portfolio	> 7d = Warning	Good
Cohort Floors (days)	904 / 1,269 / 1,634	—	Hardcoded in config
Unit Tests Passing	28 / 28	—	Complete
Dashboard Tabs	4	—	Complete
Chart Factory Functions	10	—	Complete

6 Strategic Recommendations

R1 — Upgrade Ship Mode on Interior and Pacific Routes for High-Value Orders. Interior and Pacific routes consistently show the highest delay frequencies and lowest efficiency scores. For orders carrying high-revenue products (Wonka Bar variants, Everlasting Gobstopper), logistics planners

should systematically select First Class or Same Day modes rather than defaulting to Standard Class. The incremental mode cost is materially smaller than the revenue-at-risk from delayed high-value shipments on these routes. *Target: reduce Interior and Pacific delay frequency below 30% within 6 months.*

R2 — Pursue Secondary Carrier Contracts in High-Delay States. State-level choropleth and bar chart analysis identifies specific states where delays concentrate structurally, independent of ship mode choice. Nassau Candy should negotiate contracts with secondary regional carriers in the highest-delay states to break the carrier density ceiling. Where secondary coverage is unavailable, regional inventory pre-positioning near high-delay delivery zones should be evaluated. *Target: reduce average relative lead time in the bottom-quartile states by 2 days within 12 months.*

R3 — Reassign Cross-Country Orders Where Product Specialisation Permits. Lot's O' Nuts (Arizona) and Sugar Shack (Minnesota) serve eastern-bound orders at a structural geographic disadvantage. For any product lines where cross-factory production becomes feasible in the future, routing eastern-bound orders to Georgia or Tennessee factories would yield an immediate improvement in efficiency scores on those routes. Where product specialisation prevents reassignment, the structural lead time premium should be reflected in customer-facing delivery estimates. *Target: reduce average relative lead time on cross-country routes by 2 days within 9 months through carrier-side improvements.*

R4 — Implement Revenue-Weighted Delay Alerting in the Dashboard. The current dashboard treats all delayed orders equivalently regardless of order value. A revenue-weighted alert layer should escalate same-day notifications to account managers when delays are detected on routes carrying Wonka Bar or Everlasting Gobstopper orders, which represent the highest revenue-at-risk concentration. This requires adding a revenue-threshold parameter to `config.yaml` and a new alert module in `src/analytics/kpis.py`. *Target: reduce revenue at risk from delayed high-value orders by 40% within 12 months.*

R5 — Standardise Logistics Planning Around the Dashboard. The Streamlit dashboard eliminates the dependency on ad-hoc data queries for logistics decision-making. All weekly logistics planning reviews should be conducted from live dashboard outputs filtered to the relevant factory, region, or product scope. The delay threshold slider should be calibrated to each factory's service-level agreement to ensure that threshold-based alerts are operationally meaningful rather than generic. *Target: full logistics team adoption within 3 months of deployment.*

7 Conclusion

This project has delivered a modular, production-grade logistics analytics platform for Nassau Candy Distributor's factory-to-customer shipping network. The principal contributions are:

- A cohort-floor normalisation methodology that resolves the structural 900–1,640 day lead-time artifact, isolating the 0–11 day efficiency signal that is attributable to route and ship mode choice and therefore operationally actionable.
- A five-KPI framework — average lead time, relative lead time, route efficiency score, delay frequency, and revenue at risk — with defined warning thresholds that translate route-level statistics into actionable decisions.
- A four-tab Streamlit dashboard backed by 10 Plotly chart factory functions, a typed configuration singleton, and 28 passing unit tests, meeting production quality standards for a continuously operated analytics tool.
- Route-level findings showing greater than 2× efficiency score spread across the portfolio, a consistent Same Day > First Class > Second Class > Standard Class lead time hierarchy, geographic lead time concentration in Interior and Pacific states, and structural efficiency advantages for eastern factories on eastern-bound routes.

The five strategic recommendations provide an evidence-based action framework with quantified targets. Implemented collectively, they have the potential to reduce Interior and Pacific delay frequency below the 30% warning threshold, decrease revenue at risk from high-value delayed orders by 40%, and transition logistics decision-making from reactive ad-hoc analysis to proactive, real-time, route-level intelligence.

Future work should incorporate transportation cost data to enable cost-weighted efficiency scoring, extend geographic coverage to the 200 Canadian orders currently excluded from choropleth visualisations, and integrate time-series forecasting to predict delay risk by route and month before orders are placed.

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